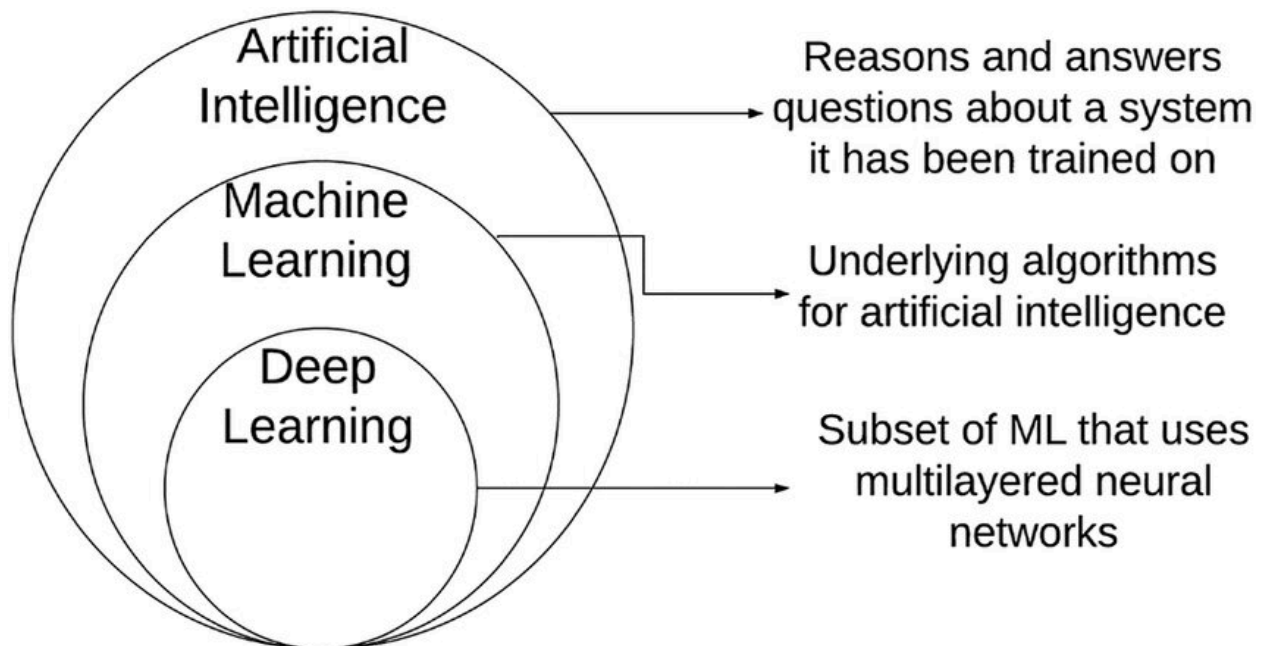


Introduction to Machine Learning

Q. What is Machine Learning?

Machine Learning (ML) is a field of study that enables computers to learn from data without being explicitly programmed. Instead of hard-coded rules, ML systems learn patterns from examples.



Classical Programming vs. Machine Learning

Classical Programming	Machine Learning
Rules are manually coded	Rules are learned from data
Deterministic behavior	Probabilistic behavior
Needs exact logic	Can generalize from examples



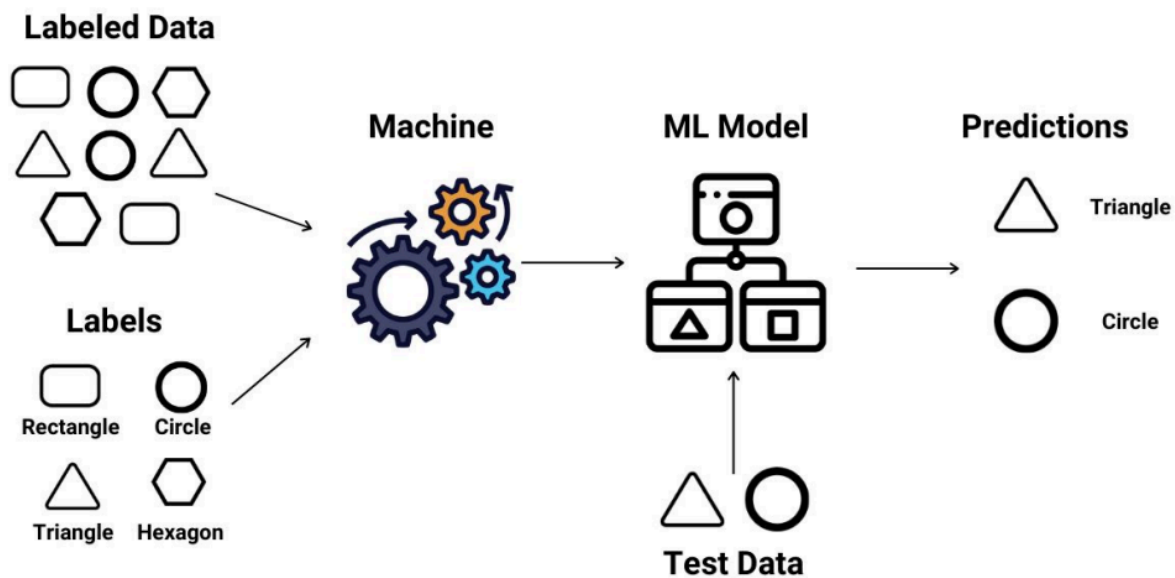
Types of Machine Learning Algorithms

1. Supervised Learning

- Learns from labeled data (inputs with corresponding correct outputs).
- The model is trained to map inputs to known outputs.
- Right answers (called ground truth) are given.

Two Main Categories:

- Classification Algorithms
- Regression Algorithms



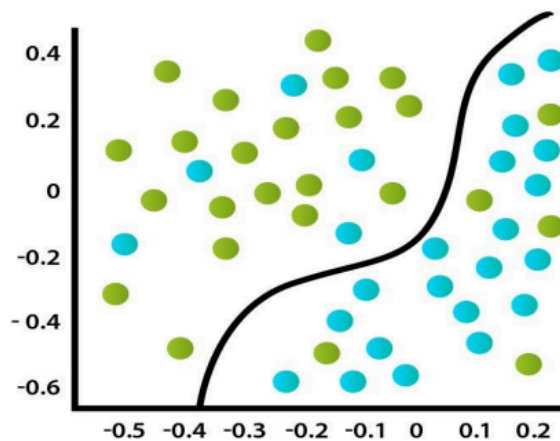
A. Supervised Learning: Classification

- Used to predict categorical labels (e.g., spam or not spam).
- Output is a discrete value.

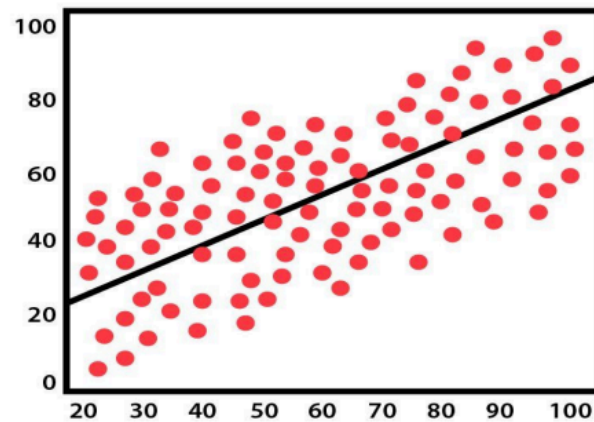
- Examples:
 - Spam Detection: Email → "spam" or "not spam"
 - Medical Diagnosis: Tumor → "benign" or "malignant"
 - Image Classification: Picture → "cat" or "dog"

B. Supervised Learning: Regression

- Predicts a continuous value from input features.
- Output is a real number.
- Examples:
 - House Price Prediction
 - Stock Price Forecasting
 - Sales Forecasting



CLASSIFICATION



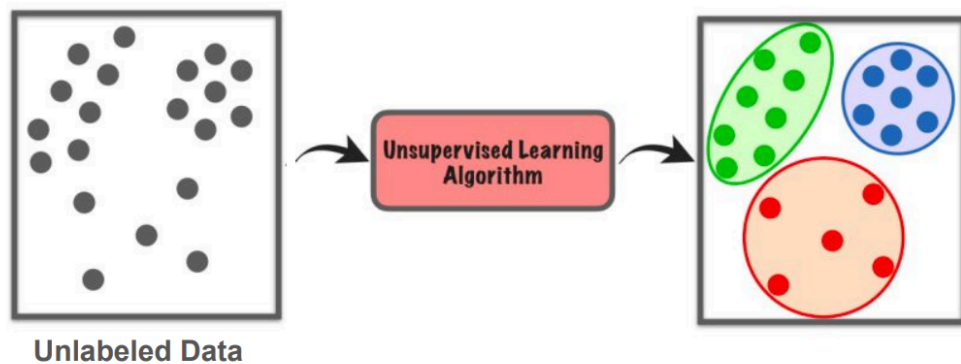
REGRESSION

2. Unsupervised Learning

- Learns from unlabeled data — no output labels are provided.
- Goal: Discover patterns, structures, or groupings in data.
- No guidance or correction is given.

Common Approaches:

- Clustering (e.g., grouping customers by buying behavior)
- Dimensionality Reduction (e.g., PCA for data visualization)

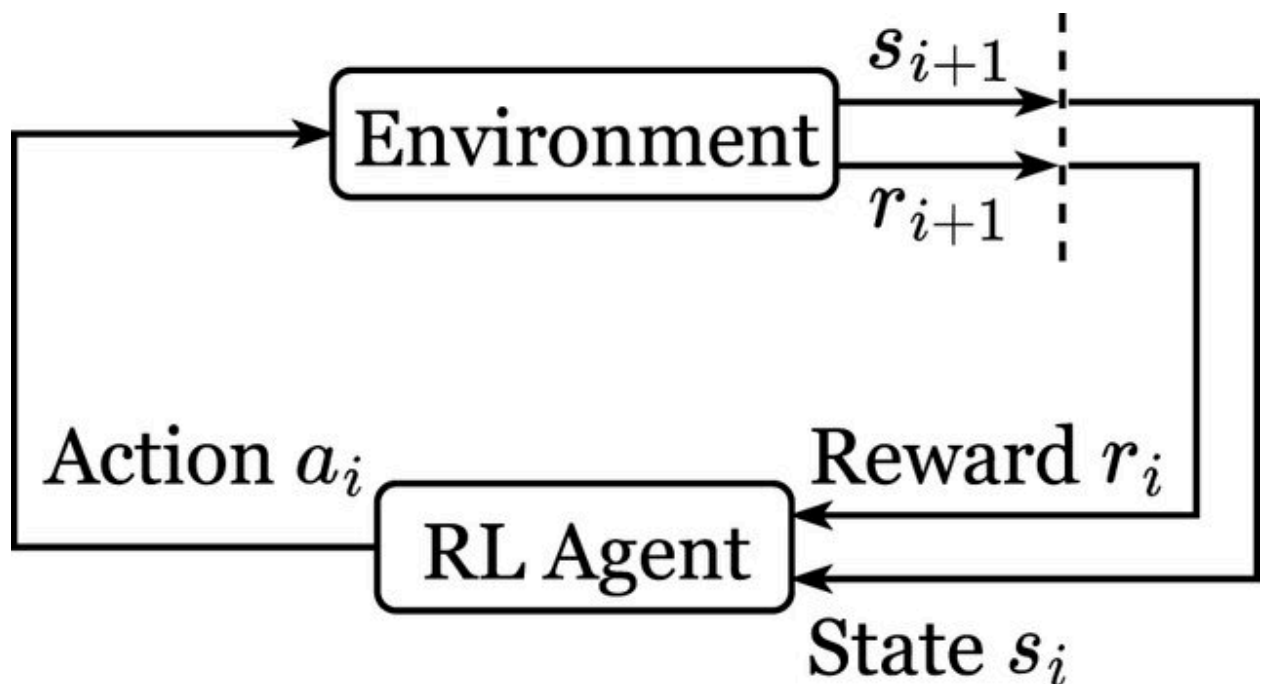


3. Reinforcement Learning

- Reinforcement Learning is a type of machine learning where an agent learns by interacting with an environment.
- Instead of labeled examples, it uses rewards (or penalties) as feedback signals.
- Goal: Learn a policy that maximizes cumulative reward over time.
- No explicit “right answer” is given—only scalar feedback after each action.

Common Approaches:

- Q-Learning (learning action-value functions)
- Policy Gradients (directly optimizing the policy)
- Deep Reinforcement Learning (using neural networks for complex state/action spaces)



Applications of Machine Learning

- **Image & Video Analysis** – e.g. object detection, medical imaging diagnostics
- **Natural Language Processing** – e.g. chatbots, sentiment analysis
- **Recommendation Systems** – e-commerce and streaming content suggestions
- **Fraud Detection** – e.g. credit-card monitoring, anomaly detection
- **Predictive Maintenance** – e.g. equipment failure forecasts in manufacturing
- **Healthcare Diagnostics** – e.g. disease prediction, personalized treatment planning
- **Autonomous Systems** – e.g. self-driving vehicles, robotics navigation
- **Customer Segmentation & Marketing** – e.g. dynamic pricing, lead scoring

Key ML Concepts (Models, Training, Features, Labels)

Model

- A mathematical function or algorithm that maps inputs to outputs
- Learns patterns from data to make predictions

Training

- The process of feeding data to a model
- Adjusts model parameters (e.g. weights) to minimize error

Features

- Individual measurable properties or attributes of input data
- Examples: pixel values in images, age and income in tabular data

Labels

- The “right answers” or target values the model tries to predict
- Examples: “spam” vs. “not spam,” price of a house

Linear Algebra Essentials

1. Linear Algebra

The study of vectors and vector spaces, and the linear maps (transformations) between them.

- **Core Objects:**
 - Scalars: single numbers (magnitudes)
 - Vectors: ordered lists of numbers or “arrows” in space
 - Matrices: 2D arrays encoding linear transformations
 - Tensors: multi-dimensional generalizations of vectors/matrices
- **Why It Matters in ML:**
 - Data points $\leftrightarrow$ vectors in high-dimensional feature spaces
 - Key algorithms (PCA, linear regression, neural layers) rely on vector/matrix operations

2. Scalars, Vectors, Matrices & Tensors

- **Scalars**
 - Elements of a field (e.g. real numbers).
- **Vectors (three views)**
 - Physics: arrows with length & direction.
 - Computer Science: ordered tuples $[v_1, v_2, \dots, v_n]$.
 - Mathematics: abstract elements of a vector space, closed under addition & scalar multiplication.
 - **Coordinate Representation**
 - 2D: (x, y)
 - 3D: (x, y, z)
- **Matrices**
 - 2-dimensional arrays
 - Used to transform or manipulate vectors, such as rotating, scaling, or reflecting them.

- **Tensors**

- Higher-order arrays (e.g. 3-way tensor for RGB images), generalizing vectors/matrices.

3. Vector Operations

- **Addition**

- Algebraic: $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

- **Scalar Multiplication**

- $\alpha(x, y) = (\alpha x, \alpha y)$: stretch/shrink by α ; flip if $\alpha < 0$.

- **Dot Product**

- Algebraic: $u \cdot v = \sum_i u_i v_i$
- Geometric: $|u| |v| \cos \theta$
- Measures projection of one vector onto another; zero if perpendicular.

- **Cross Product**

- Algebraic: $u \times v = (u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1)$
- Geometric: $|u| |v| \sin \theta$
- Measures the area of the parallelogram spanned by u and v ; zero if vectors are parallel.

- **Linear Combination**

- Any vector v can be written as linear combination of standard unit vector (i and j) such that $v = a i + b j$.
- Span: all combos of a set of vectors.

- **Linear Dependence vs. Independence**

- Independent: none can be written as a combo of others.
- Dependent: at least one redundant (lies in span of the rest).

4. Matrix Operations

- Addition
- Subtraction
- Scalar Multiplication
- Matrix Multiplication

- Transpose

5. Determinant, Rank, Inverse

- **Determinant (det M)**
 - Tells how a matrix scales in space (area in 2D, volume in 3D)
 - Positive: shape is preserved
 - Negative: shape is flipped
 - Zero: space is squashed flat → no inverse
- **Rank**
 - Number of independent columns in a matrix
 - Tells us how much of the space the matrix covers
 - Full rank: keeps all dimensions
 - Low rank: compresses or flattens the space
- **Inverse Matrix (M^{-1})**
 - Exists if and only if $\det M \neq 0$.
 - To “Undo” transformation: $M^{-1}M = I$.

6. Eigenvalues & Eigenvectors

- A matrix transforms vectors (moves, stretches, rotates them).
- Eigenvectors are special vectors that don't change direction when a matrix is applied, they only get stretched or squished.
- The amount they are stretched or squished is called the eigenvalue.
- If a matrix M acts on a vector v , and the result is just a scaled version of v , like this:

$$Mv = \lambda v$$

- v is an eigenvector
- λ is its eigenvalue

7. Cramer's Rule

- Cramer's Rule is a method to solve a system of linear equations using determinants.
- It works for systems with as many equations as unknowns (e.g., 2 equations, 2 variables).

Q. How does it work (for 2 variables) ?

For the system:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Let:

- D = determinant of the coefficient matrix
 $D = a_1b_2 - a_2b_1$
- Replace the x -column with constants to get D_x
- Replace the y -column with constants to get D_y

Then: $x = D_x/D, y = D_y/D$

Probability Fundamentals

Random Variables (RVs)

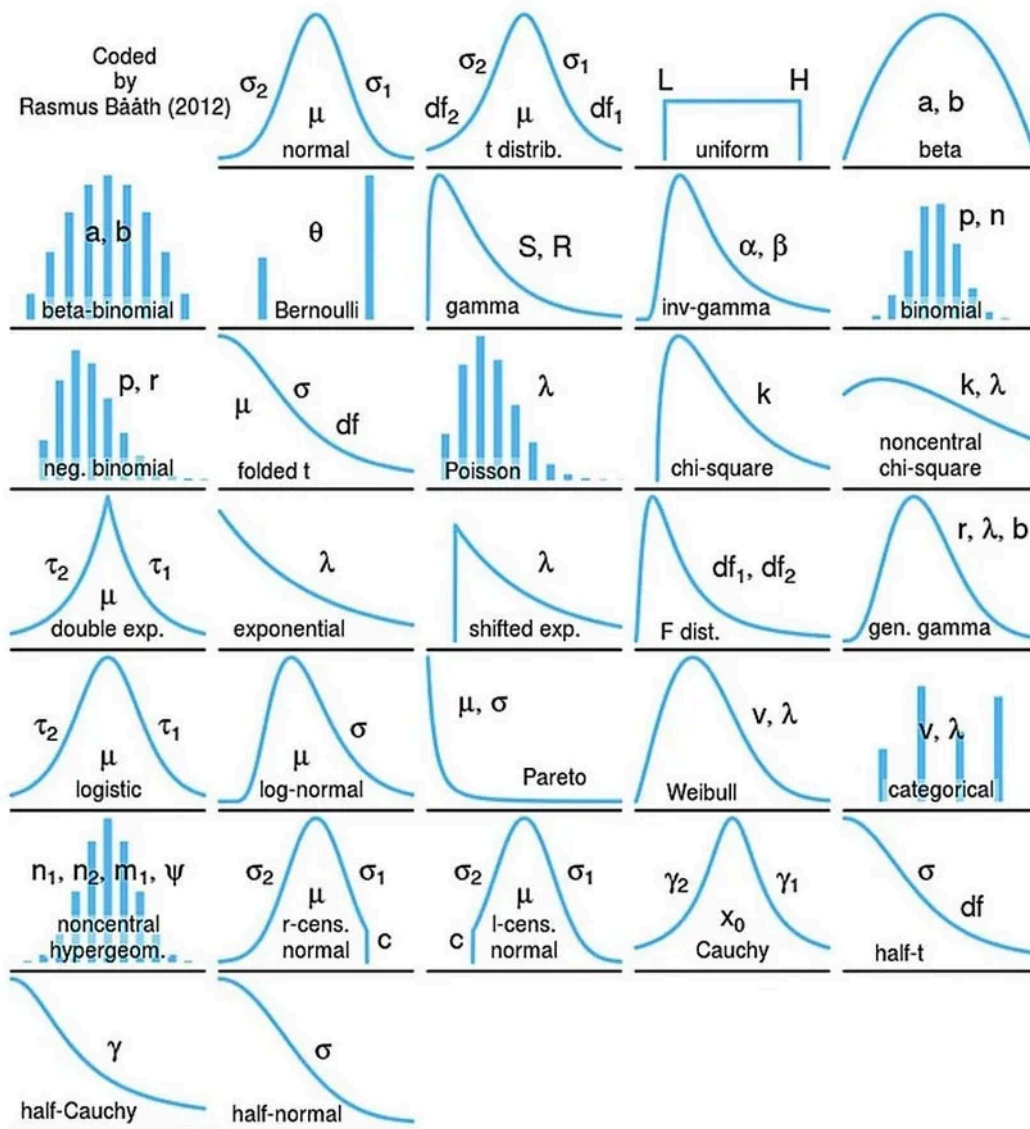
A Random Variable is a function that is used to describe the outcomes of a random experiment or process. RV in statistics and probability theory has uncertain or probabilistic values determined by the outcomes of a random experiment or process.

Type	Description	Example
Discrete	Takes countable values	Number of heads in 5 coin flips
Continuous	Takes values from a range of real numbers	Height of a person, time to failure

Probability Distribution

A probability distribution shows all possible outcomes of a random experiment with their probabilities. Examples: Toss a coin $\rightarrow$ {Heads: $\frac{1}{2}$, Tails: $\frac{1}{2}$ }

Famous Distributions



Normal – Bell-shaped curve

Poisson – Rare events over time

Uniform – All outcomes equally likely

Exponential – Time between events

Binomial – Discrete success/failure trials

Log-normal – Positively skewed

Chi-square – Used in hypothesis testing

Pareto – Power-law (80/20 rule)

Beta – Values between 0 and 1 (flexible shape)

Importance of Probability Distribution

- Shows how data behaves
- Helps in decision-making
- Useful in predictions, modeling, finance, science, and more

Probability Mass Function (PMF)

The PMF assigns a probability to each possible value of a discrete random variable.

$$P(X=x) = \text{Probability that } X \text{ takes value } x$$

Properties

- $0 \leq P(X=x) \leq 1$
- $\sum_{x \in \text{range of } X} P(X=x) = 1$

Example:

Let X be the result of a die roll.

$$P(X=x) = 1/6, x=1,2,\dots,6$$

Probability Density Function (PDF)

The PDF defines the relative likelihood of a continuous random variable falling near a value.

$$\text{For continuous variables, } P(X=x) = 0$$

Properties

- $f(x) \geq 0$
- Total area under the curve is 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Probability Over Interval

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Example (Normal Distribution)

$$f(x) = \frac{1}{\sqrt{2\pi}(\sigma^2)} \exp(-((x-\mu)^2)/2(\sigma^2))$$

Discrete vs. Continuous

Feature	Discrete	Continuous
Random Variable	Countable values	Infinite range
Probability Function	PMF	PDF
Probability at a point	$P(X=x) > 0$	$P(X=x) = 0$
CDF	$\sum_{xi \leq x} P(X=xi)$	$\int_{-\infty}^x f(t) dt$
Probability in interval	$\sum_{a \leq x \leq b} P(X=x)$	$\int_a^b f(x) dx$
Visualization	Bar graph	Smooth curve

Cumulative Distribution Function (CDF)

The CDF gives the probability that a random variable is less than or equal to a value:

$$F(x) = P(X \leq x)$$

Discrete Case

$$F(x) = \sum_{xi \leq x} P(X=xi)$$

Continuous Case

$$F(x) = \int_{-\infty}^x f(t) dt$$

Properties

- $F(x) \in [0,1]$
- Non-decreasing
- $\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1$

Density Estimation

Used to estimate the underlying probability distribution from sample data.

Parametric Estimation

Assumes data follows a known distribution (e.g., Gaussian).

Steps:

1. Choose a distribution model (e.g., normal, exponential).
2. Estimate parameters (mean μ , variance σ^2 , etc.).

Pros:

- Easy to interpret
- Requires fewer data points

Cons:

- Assumption sensitive: Wrong model $\rightarrow$ wrong results

Non-Parametric Estimation

No assumption about the data's distribution shape.

Methods:

- Histogram: Bins data, simple but coarse.
- KDE (Kernel Density Estimation):

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K(x - x_i)/h$$
 - K: Kernel function (e.g., Gaussian)

- h : Bandwidth (smoothing parameter)

Pros:

- More flexible
- Adapts to data shape

Cons:

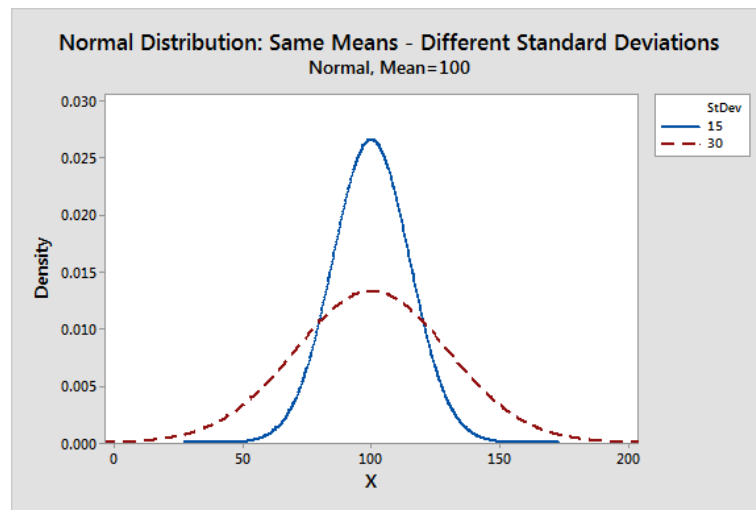
- Sensitive to bandwidth choice
- Requires more data

Normal (Gaussian) Distribution

The Normal Distribution (also called Gaussian distribution) is a continuous probability distribution used to describe real-valued variables that cluster around a mean.

Key Characteristics:

- Bell-shaped curve: Symmetric around the mean (μ).
- Spread of the curve is measured by standard deviation (σ).
 - Smaller $\sigma \rightarrow$ Tall, narrow curve
 - Larger $\sigma \rightarrow$ Short, wide curve



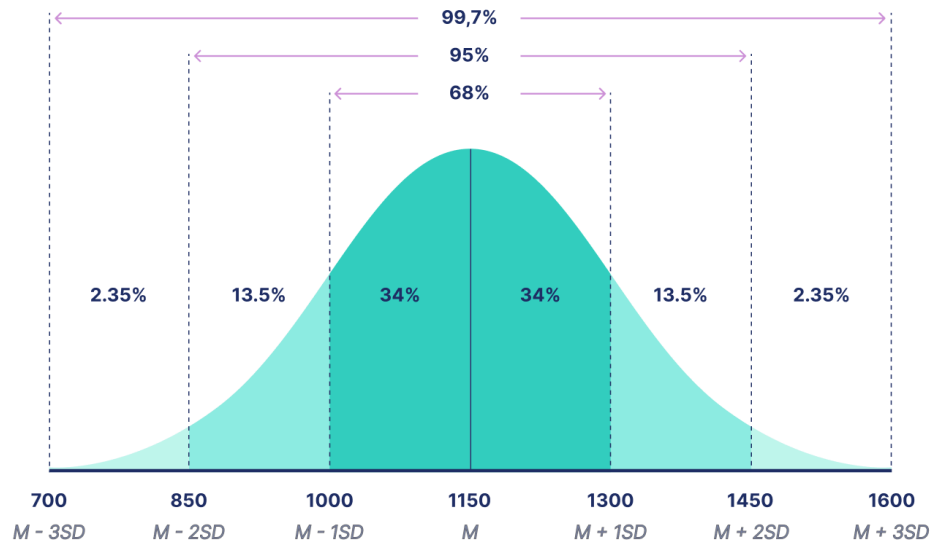
- Mean (μ) determines the center of the distribution.

Empirical Rule (68-95-99.7 Rule):

For a normal distribution:

- 68% of data lies within $\pm 1\sigma$
- 95% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

Using the empirical rule in a normal distribution



Standard Normal Distribution (Z-distribution):

- Special case of the normal distribution.
- Mean = 0, Standard Deviation = 1
- Used to calculate probabilities via Z-scores and Z-tables.

Q. Why is it useful?

- Models many natural phenomena (e.g., heights, IQ scores).
- Helps calculate probabilities by finding areas under the curve.

Probability Calculations:

- Involve integration of the probability density function (Area under the curve).
- Usually done using:
 - Z-tables
 - Statistical software (e.g., Python, R, Excel)

Central Limit Theorem

The Central Limit Theorem says that when we add many independent random variables (like rolling dice), their sum or average will follow a normal (bell-shaped) distribution, even if the original variables are not normal.

Key Ideas:

- Mean & Standard Deviation:
 - The mean becomes more stable as sample size increases.
 - The spread (standard deviation) grows slowly, making the curve smoother and more predictable.
- Galton Board Example: Balls fall randomly left or right. At the bottom, they form a bell curve. This shows how random events create a predictable pattern when repeated many times.
- Dice Rolling Example: Roll dice, add the results → the distribution of the total becomes bell-shaped. This happens even if:
 - Dice are not fair
 - Outcomes are not originally normally distributed

Assumptions:

- Independent random variables
- Same distribution
- Finite mean and variance

As we add more variables, the sum gets closer to a normal curve.

Q. Why does Normal Distribution appear?

- It has a special shape: defined by the function $f(x) = \frac{1}{\sqrt{2\pi(\sigma^2)}} \exp\left(-\frac{(x-\mu)^2}{2(\sigma^2)}\right)$
- When you add two normal distributions, you get another normal distribution.

- It's the only function that stays the same shape when added — a very special property.
- This is why the normal distribution is universal when summing many variables.

Q. Why does π appear in the Normal Formula?

- The formula for the normal distribution includes π because of geometry:
 - Think of the bell curve as part of a 3D symmetrical shape
 - The math involves calculating volume under this shape, which includes π
- This connects to symmetry and circular shapes — the curve behaves the same in all directions.